ABSTRACT

LUNG CANCER DETECTION USING DEEP LEARNING ON CT SCAN IMAGES:

Lung cancer is one of the leading causes of death worldwide, and early detection is essential to improve survival rates. Conventional diagnostic methods such as CT scan analysis by radiologists may lead to delays, human errors, and variability in diagnosis. To address these challenges, this project proposes an automated deep learning-based approach for accurate and early detection of lung cancer using CT scan images.

The proposed system follows a structured pipeline that includes preprocessing, segmentation, classification, and explainability. Initially, CT images are enhanced using noise removal, contrast improvement, and data augmentation techniques. A U-Net model with a ResNet-34 backbone is used for segmenting lung regions and detecting nodules. For classification, multiple deep learning models such as ResNet-50, ResNet-101, GoogLeNet, and EfficientNet-B0 are applied using transfer learning to classify images into four categories: Adenocarcinoma, Large Cell Carcinoma, Squamous Cell Carcinoma, and Normal.

The performance of the models is evaluated using accuracy and ROC analysis, where GoogLeNet achieved the best accuracy of 91.75%. Additionally, Grad-CAM is used to generate visual heatmaps for model interpretability. The proposed system provides an efficient, accurate, and reliable solution for lung cancer detection, supporting medical professionals in faster and better decision-making.

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